



IOT-BASED SMART SHUTTLE MANAGEMENT SYSTEM FOR UNIVERSITY CAMPUS

CSE 426/CSE 4326 : Microprocessors and
Microcontrollers Laboratory
Section: (H)

Meet our team



Md. Salman Khan
011 221 319



Sumaiya Islam Babli
011 221 092



Mrs. Tonni Islam
011 221 329



MITHILA FARJANA
011 221 575



MITHUN SAHA
011 221 242



AL SHAHRIAR ABIR
011 202 338

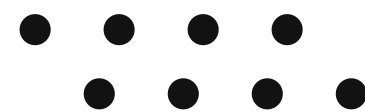


Microcontrolle: Arduino Mega

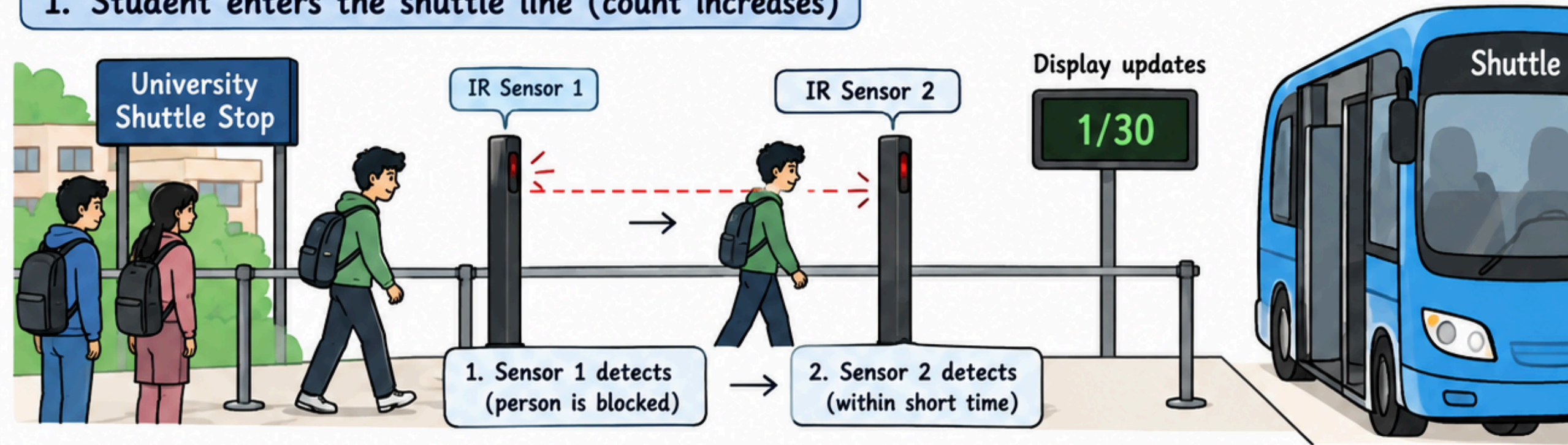




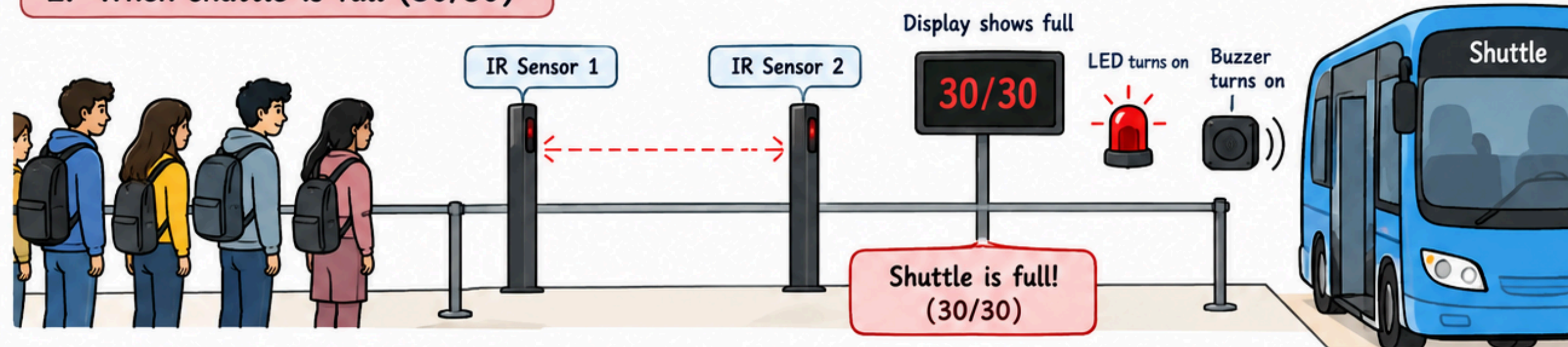
Feature (01): Shuttle Line Queue Management System



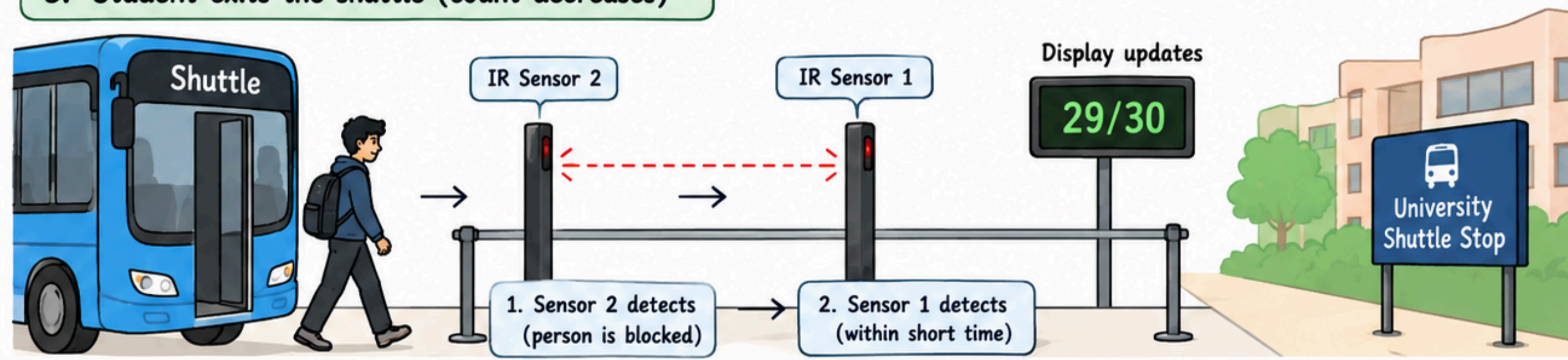
1. Student enters the shuttle line (count increases)



2. When shuttle is full (30/30)



3. Student exits the shuttle (count decreases)



IoT Based Smart Shuttle Management System for University Campus

Arduino Mega

IR Sensor 1
(Entry & Exit)

IR Sensor 2
(Entry & Exit)

Buzzer 1

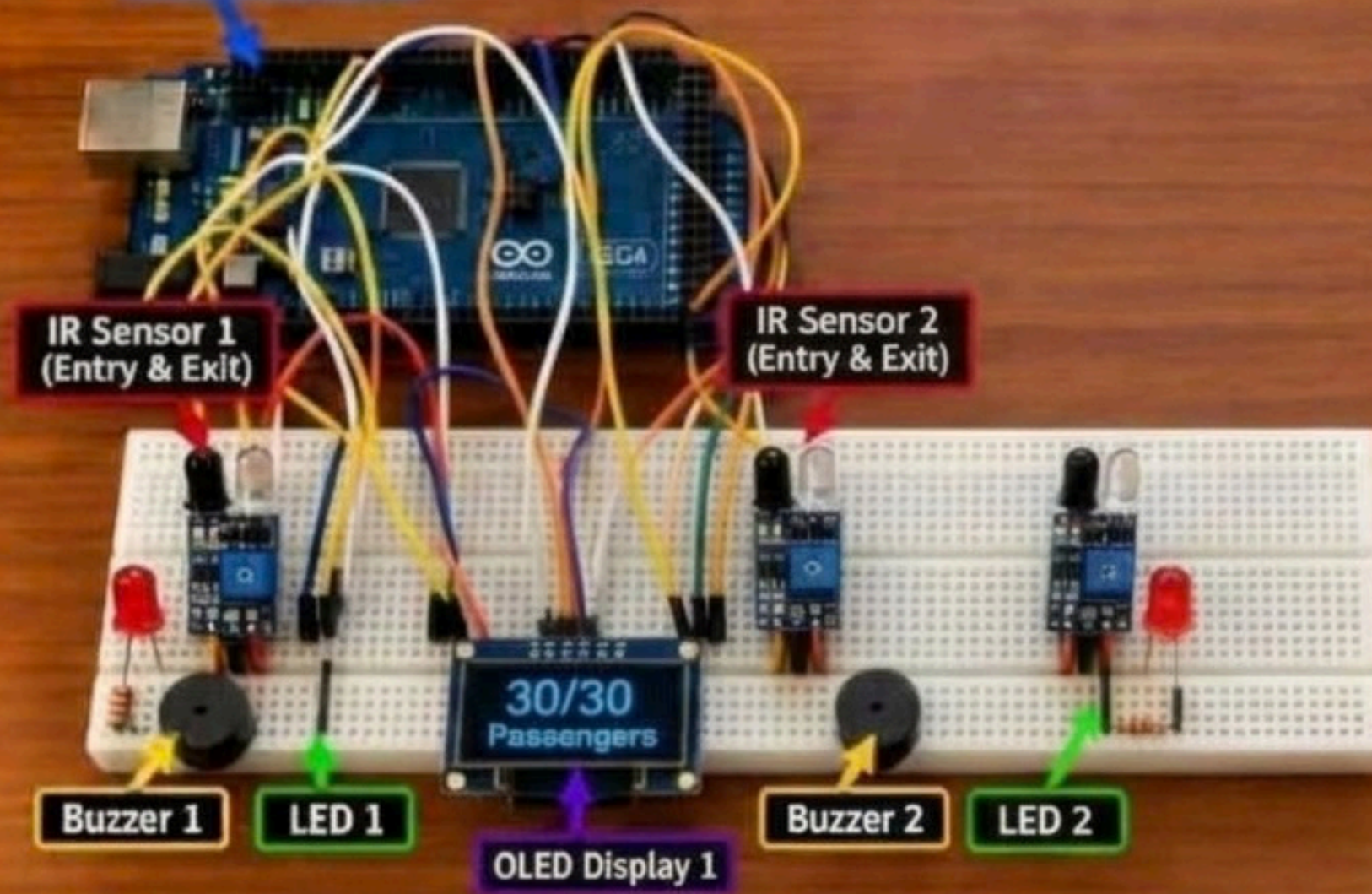
LED 1

OLED Display 1

Buzzer 2

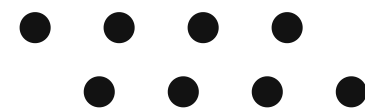
LED 2

Feature 01: Shuttle Line Queue Management System

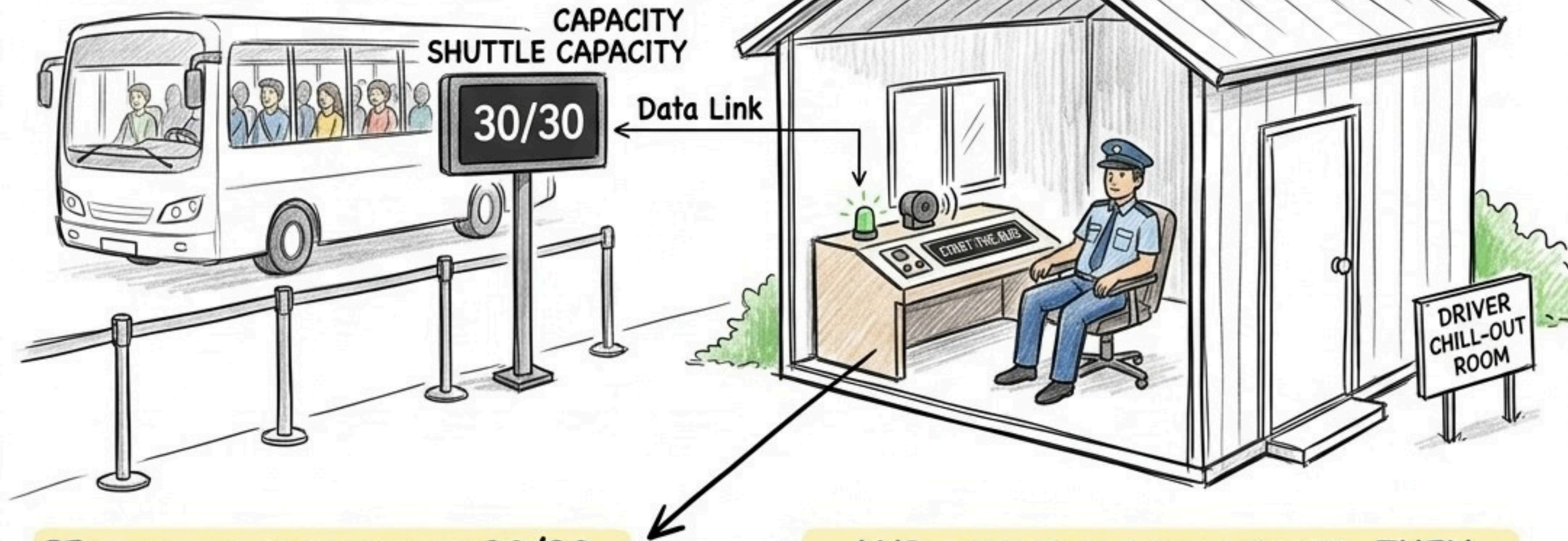




Feature (02): Driver Room Alarm System

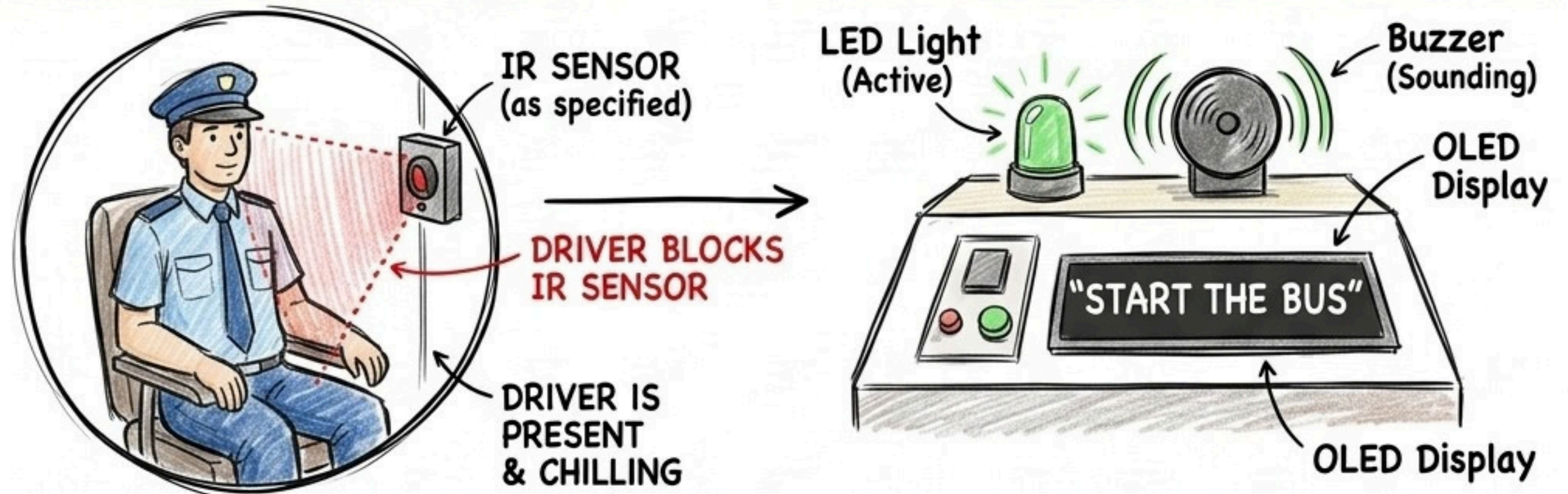


1. SHUTTLE CAPACITY REACHES FULL (30/30)
(Students fully loaded)

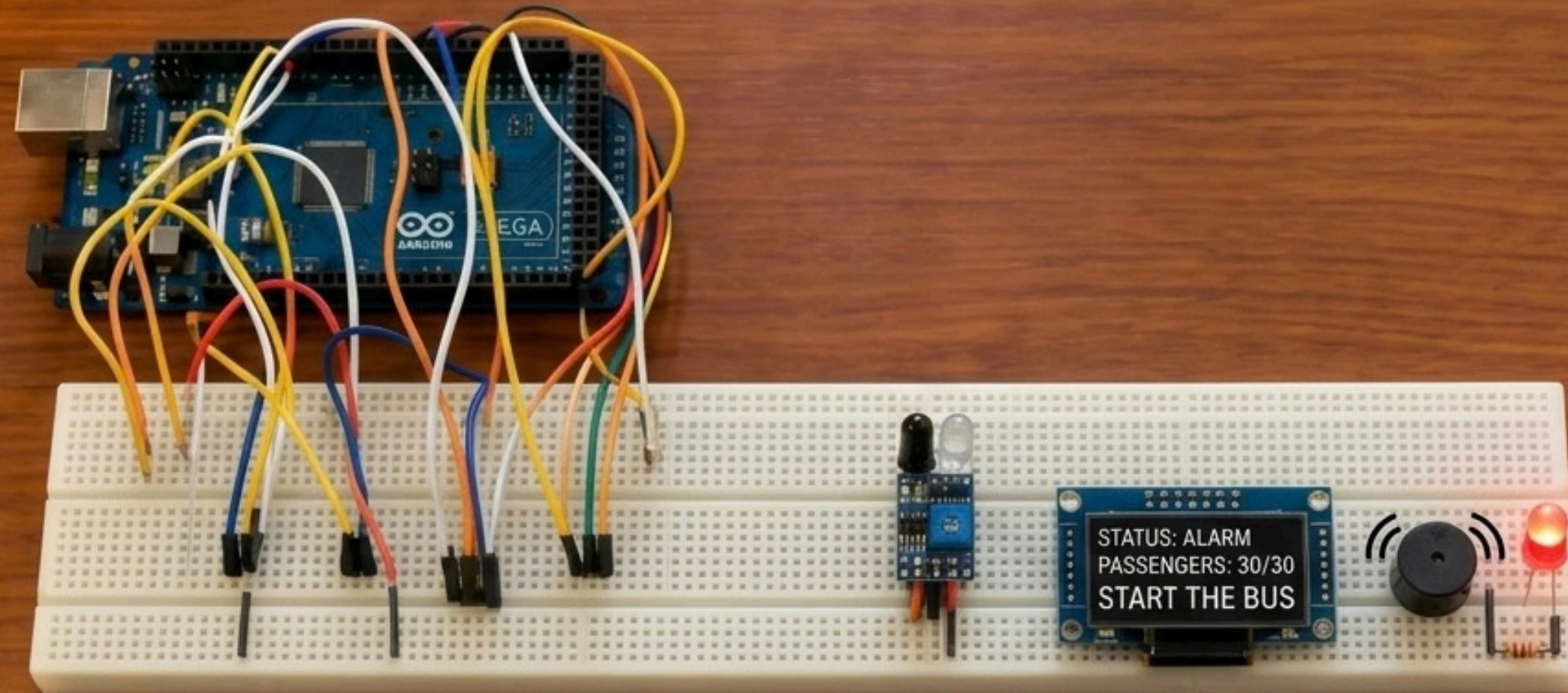


IF SHUTTLE CAPACITY IS 30/30...

...AND DRIVER BLOCKS SENSOR, THEN...



Feature (02): Driver Room Alarm System



Microcontroller: ESP32





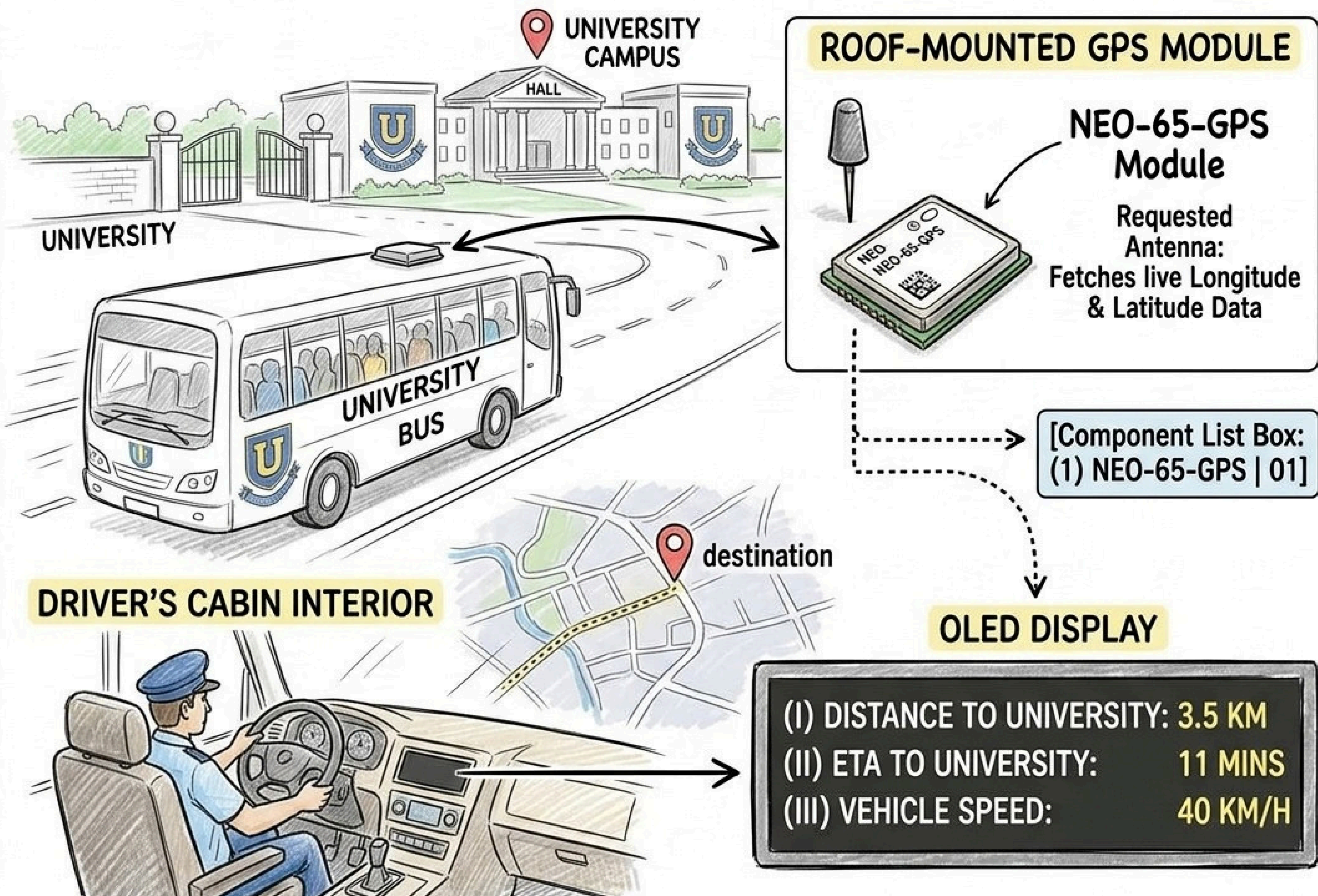
Feature (03):



Real-time Shuttle Navigation System

• • • • •





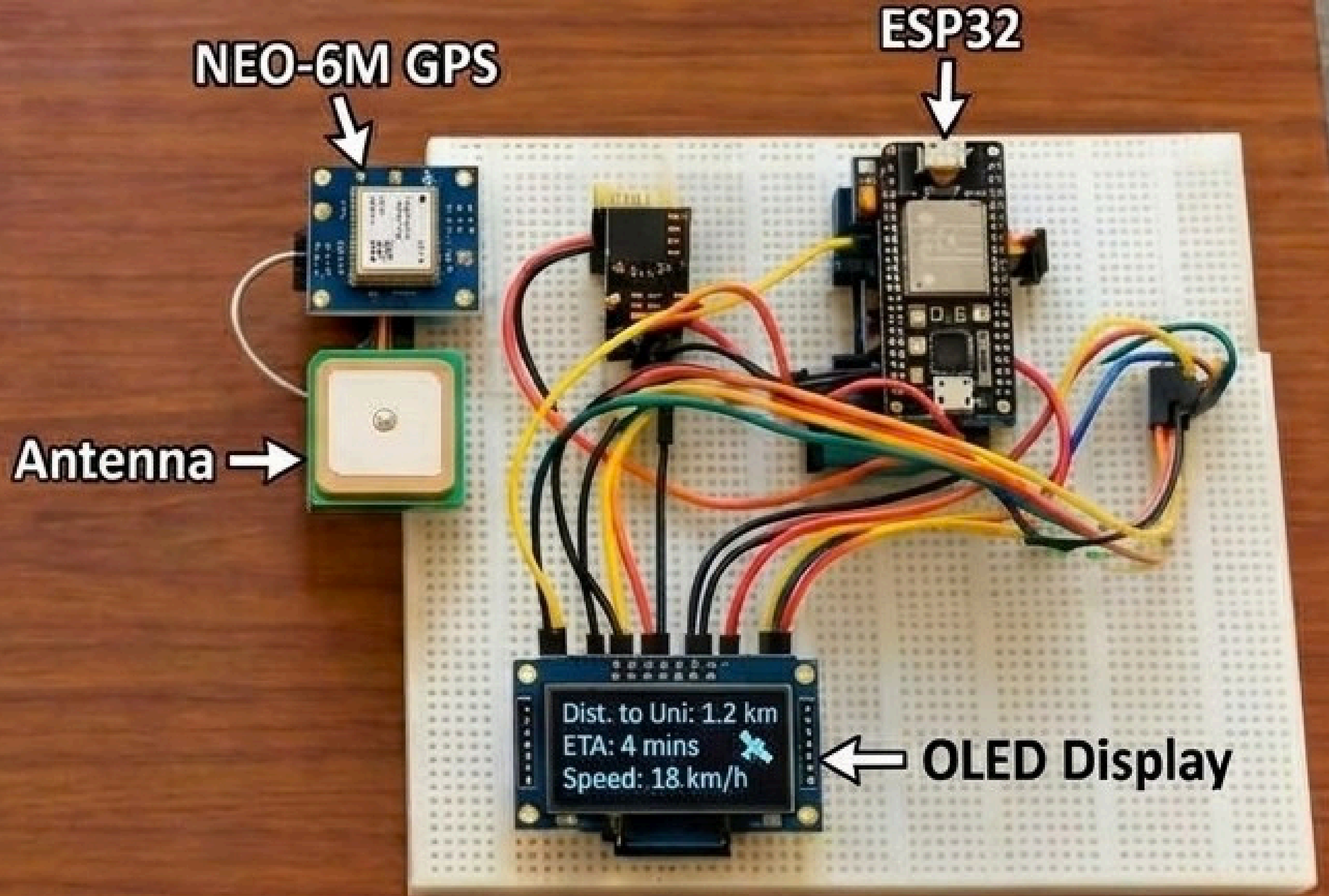
NEO-6M GPS

ESP32

Antenna

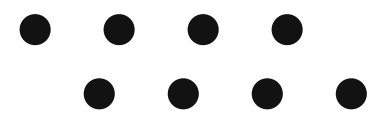
OLED Display

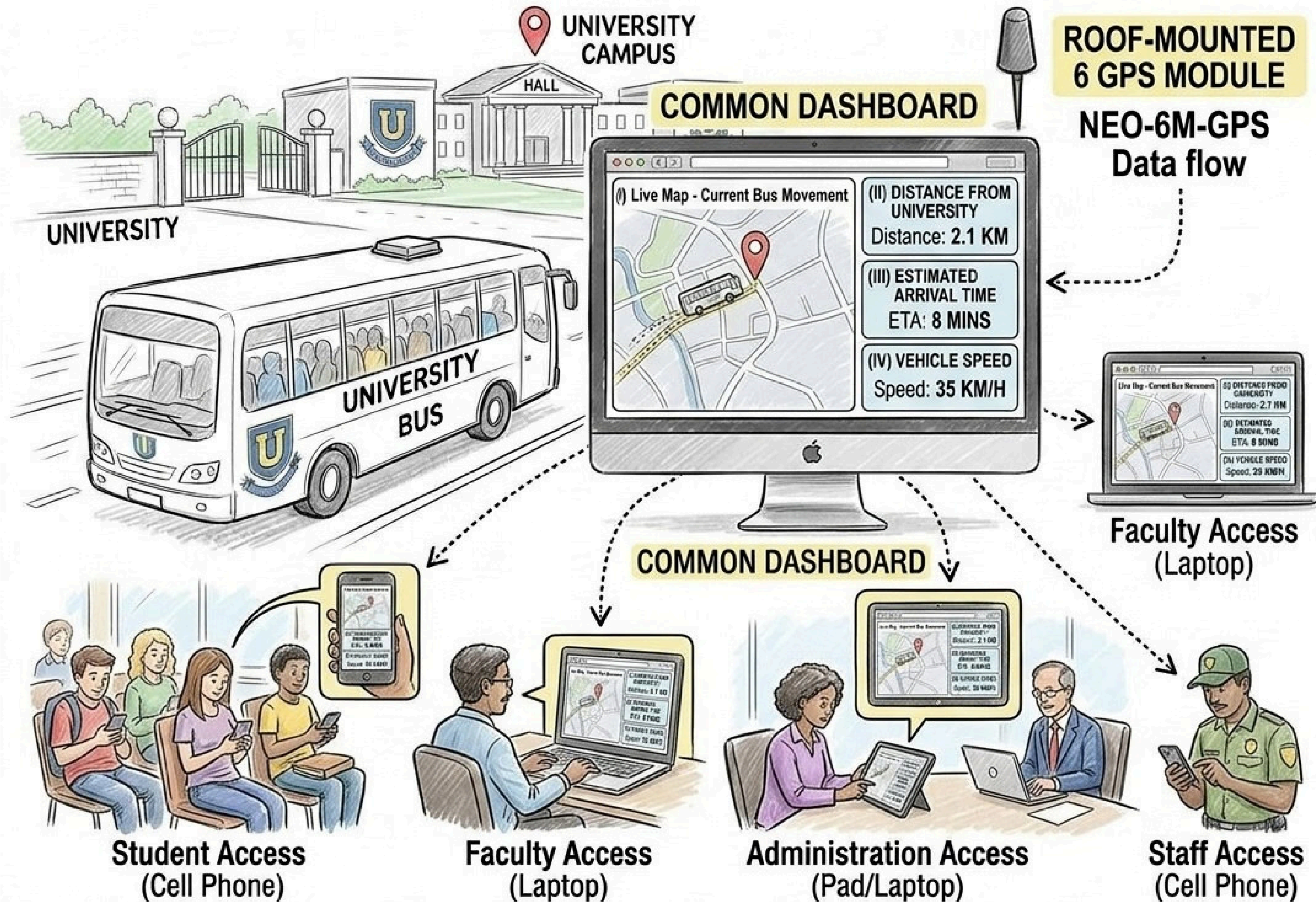
Dist. to Uni: 1.2 km
ETA: 4 mins
Speed: 18.km/h

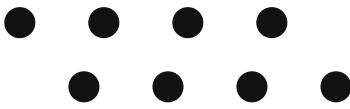
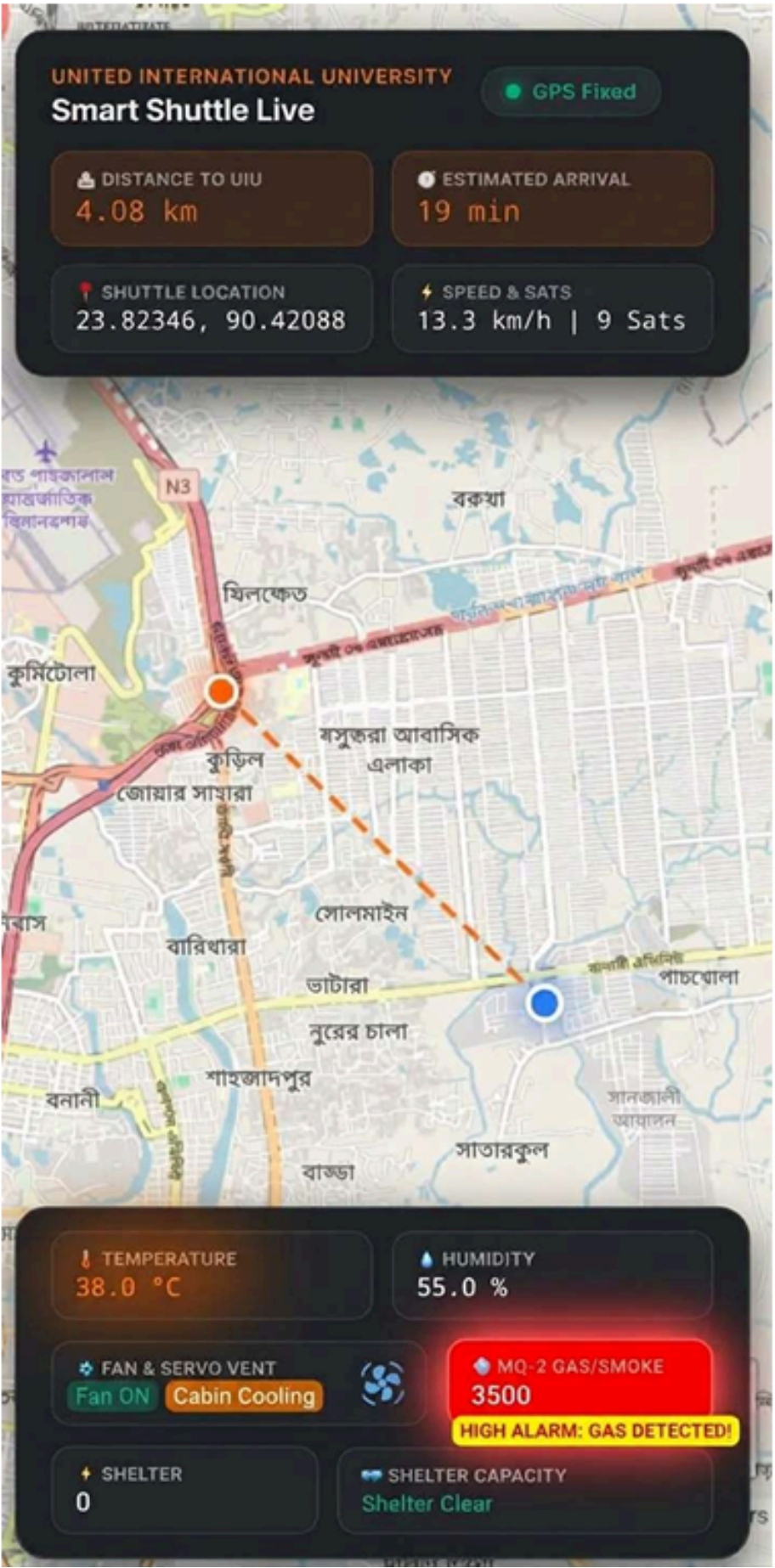
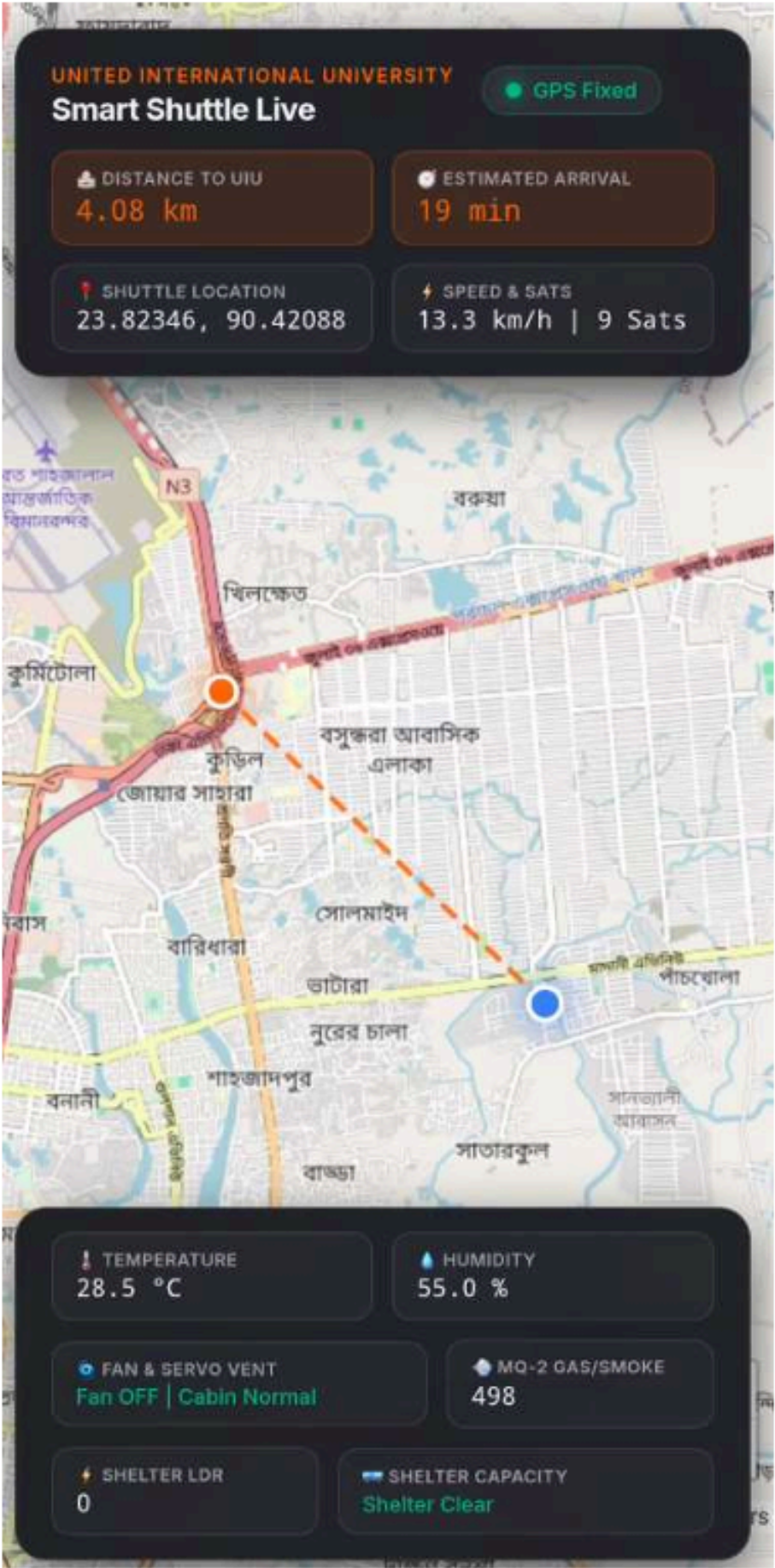




Feature (04): Live Shuttle Tracking Dashboard





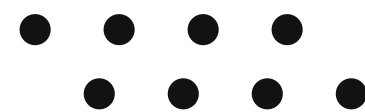


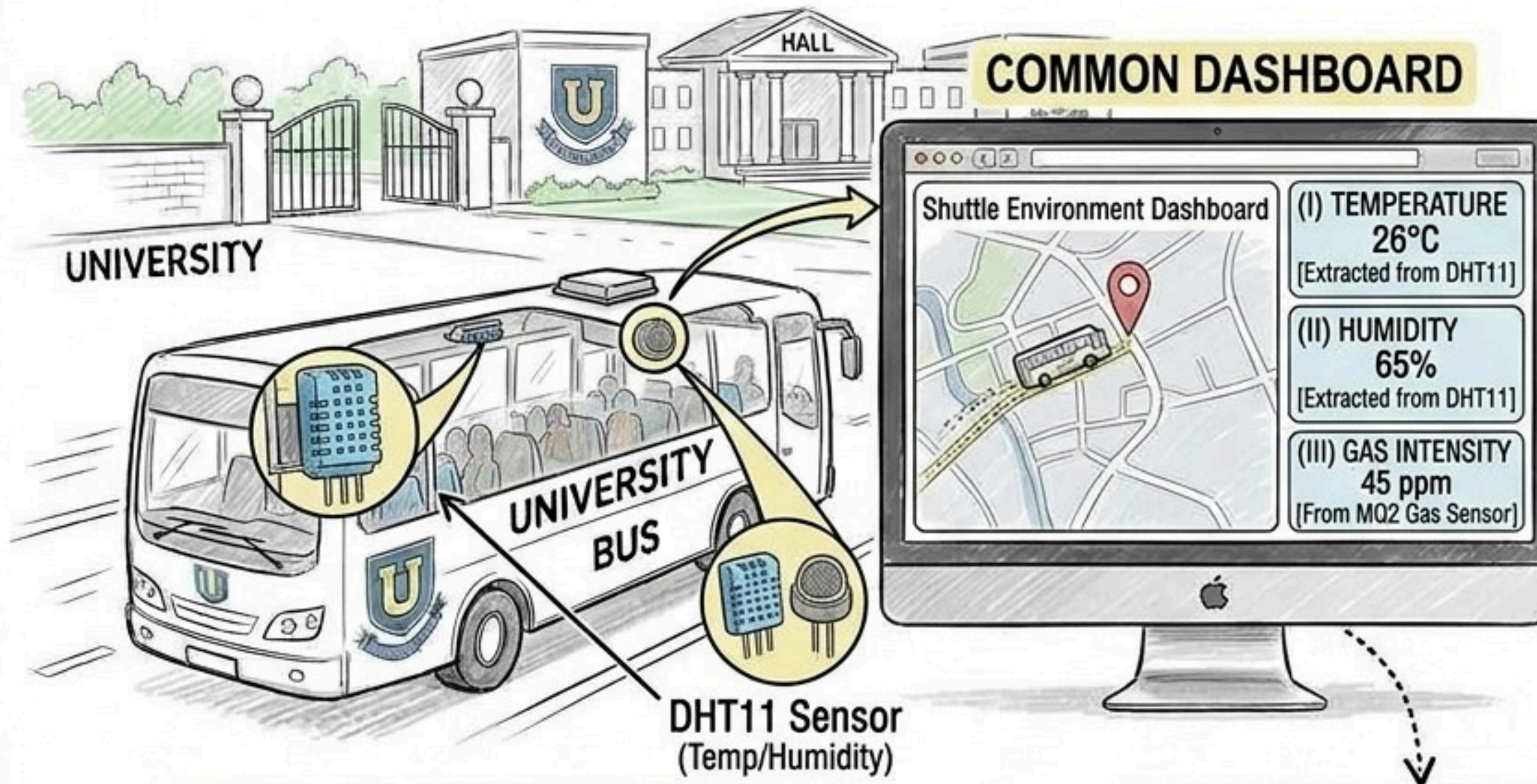


Feature (05):

Shuttle Environment

**Monitoring System, Risk assessment and alert,
Cooling system**





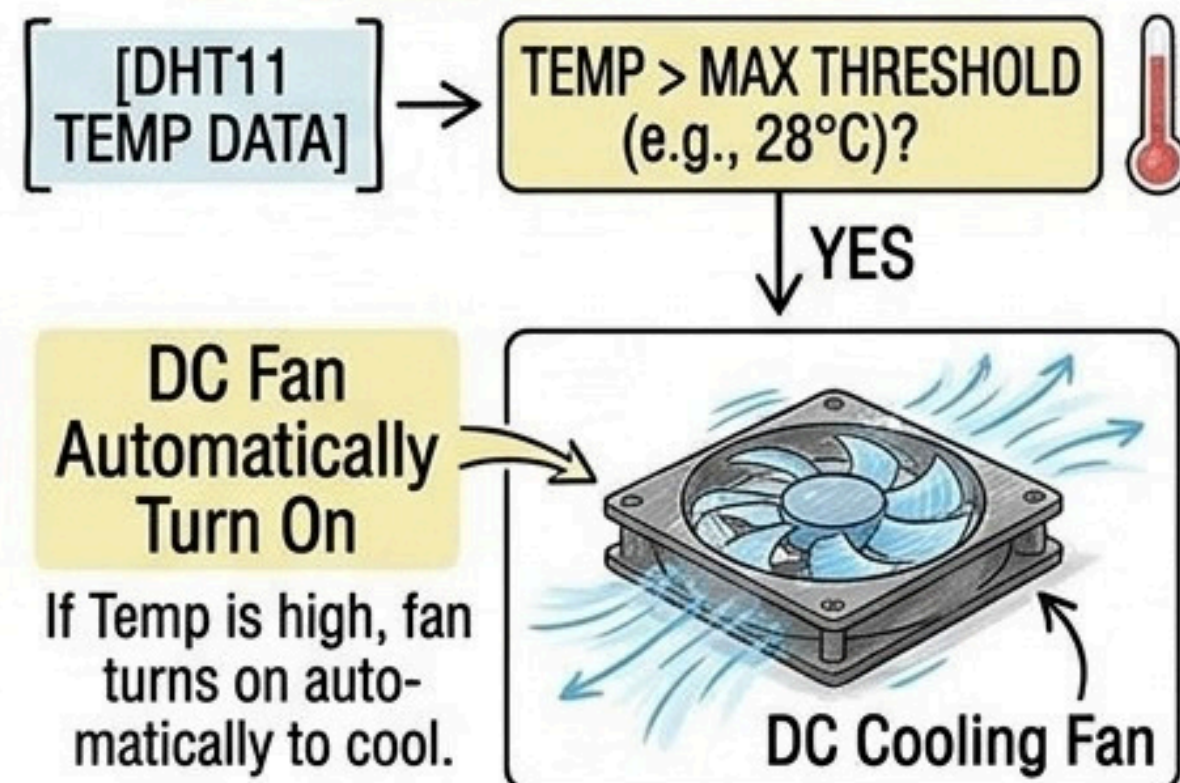
COMPONENT LIST

SENSOR	VALUE
DHT11 Sensor	DHT11
DHT11	26°C
HUMIDITY	65%
GPS Sensor	NEO-6M
GAS INTENSITY	45 ppm

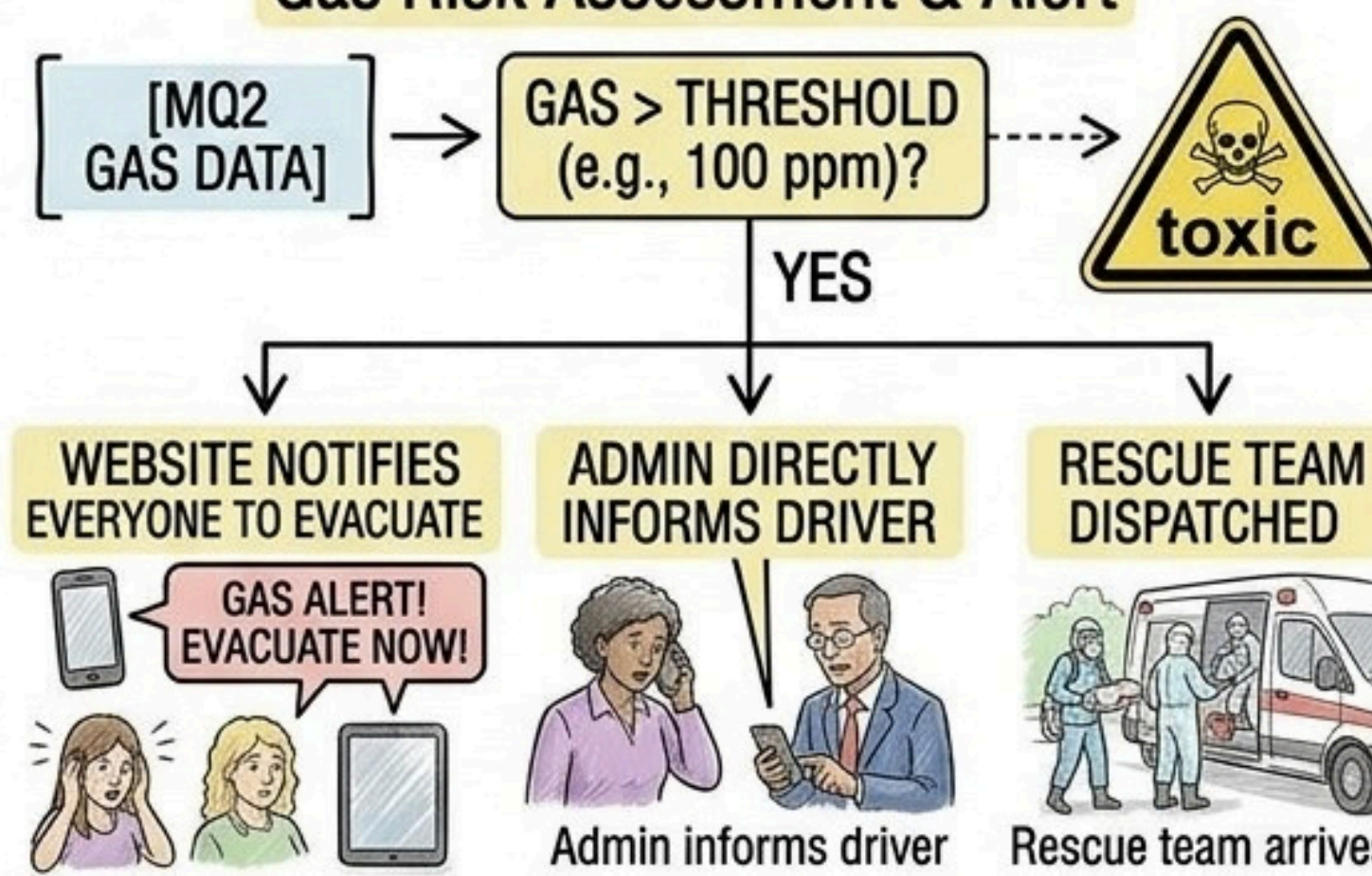
COMPONENT LIST

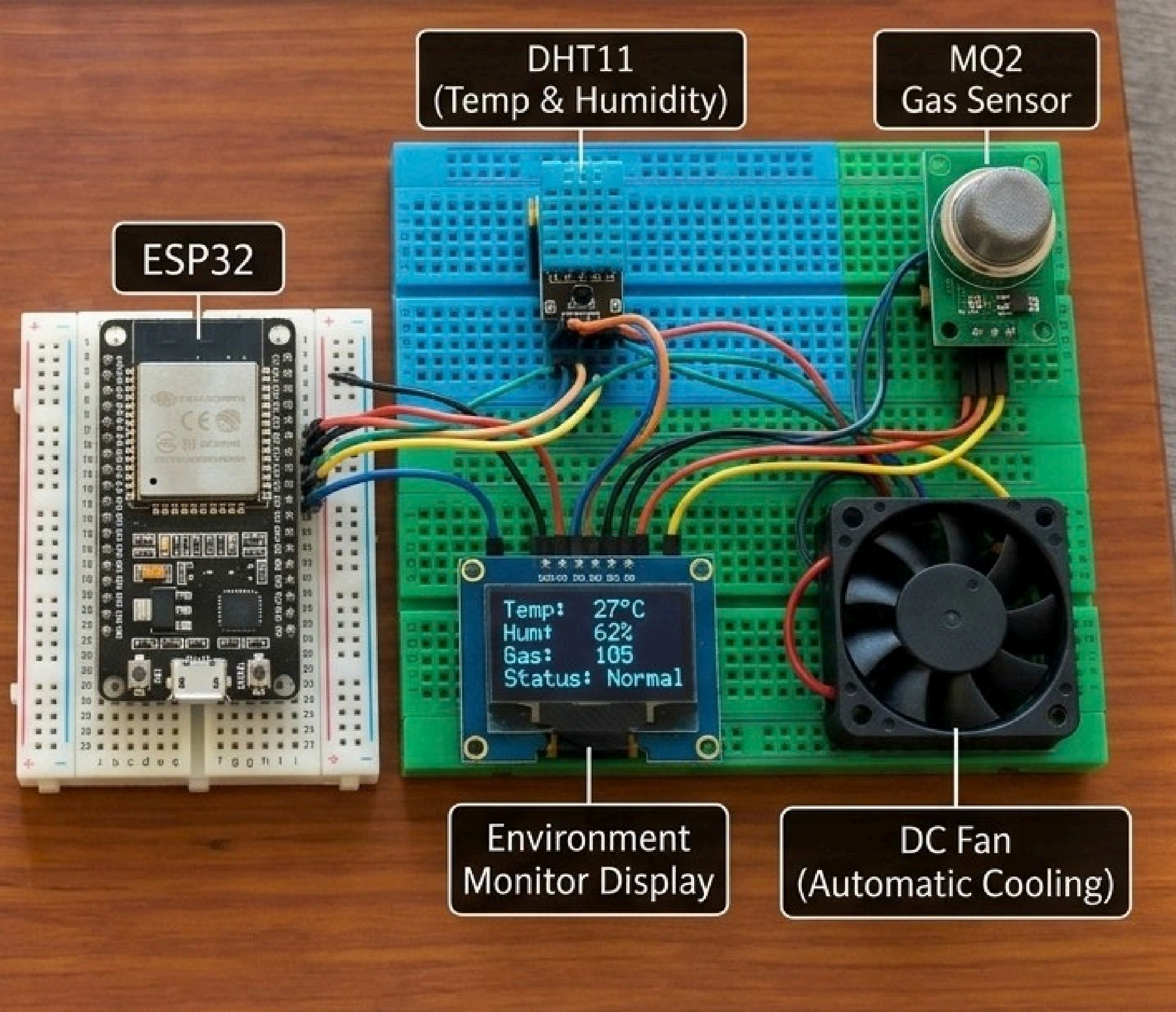
FIELD	VALUE
TEMPERATURE	26°C
HUMIDITY	65%
GAS INTENSITY	45 ppm
SPENSITY	120 ppm
STAFF	Staff

Temperature Risk Control

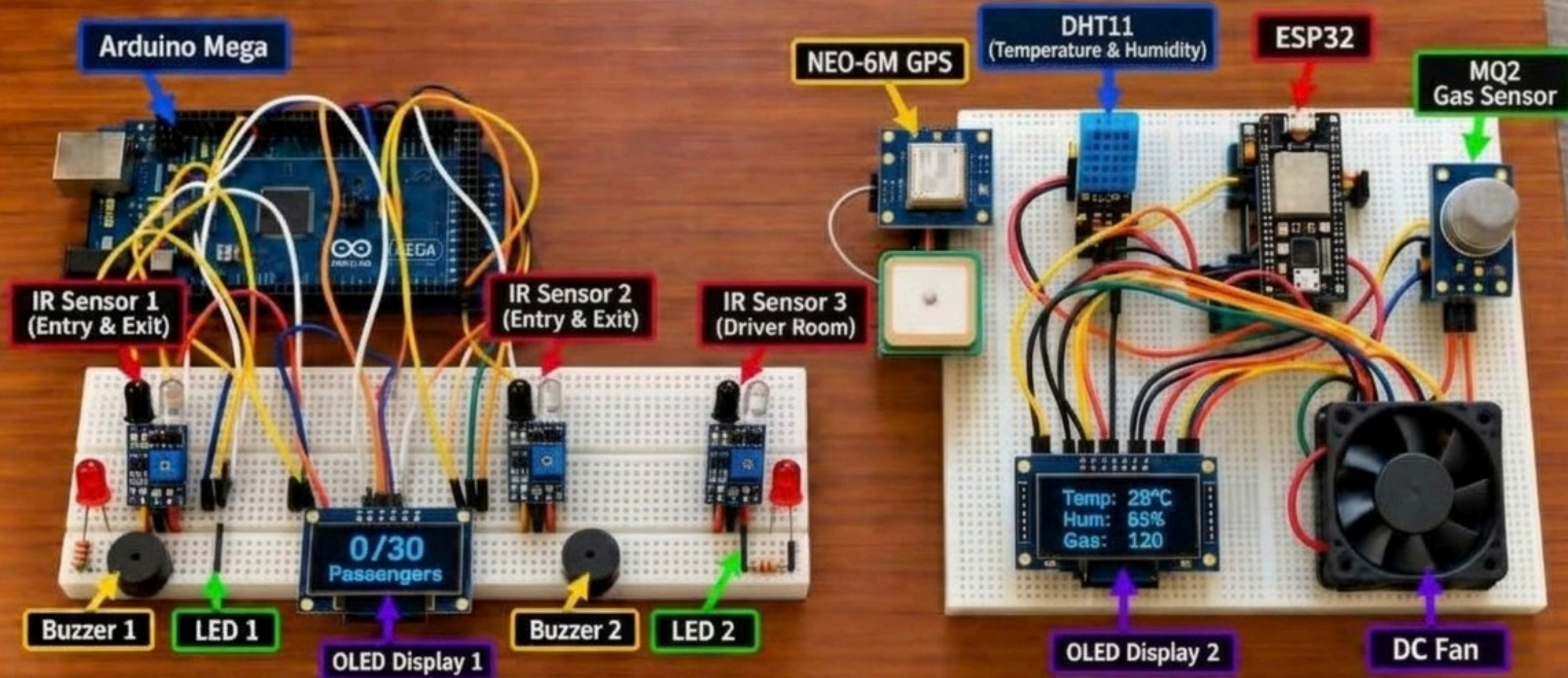


Gas Risk Assessment & Alert





IoT Based Smart Shuttle Management System for University Campus



Feature 01: Shuttle Line Queue Management System
Feature 02: Driver Room and Alarm System

Feature 03: Real-time Shuttle Navigation System
Feature 04: Live Shuttle Tracking Dashboard
Feature 05: Shuttle Environment Monitoring,
Risk Assessment, and Cooling System



Thank You

